1. Clinical Question & Core Hypothesis

Clinical Question

Why do some tumors with favorable immune architecture still fail to achieve better clinical outcomes?

Working Hypothesis

In triple-negative breast cancer (TNBC) treated with neoadjuvant chemotherapy, the predictive value of stromal TIL density (sTIL%) for pathologic complete response (pCR) is contingent on spatial organization: high sTIL% predicts higher pCR only when TILs are enriched in close proximity to tumor nests or at the tumor–stroma border. To test this, we hypothesize that (1) a composite morphological score that combines sTIL% with a proximity/organization index (e.g., fraction of TILs within defined radii of tumor nests, mean distance from TILs to tumor nests, and a TIL clustering metric) will outperform sTIL% alone in predicting pCR; (2) there will be a significant interaction between sTIL% and the proximity/organization index, such that the strongest association with pCR occurs when both high density and favorable spatial arrangement are present; (3) this morphology-based predictor will maintain predictive power across tumor heterogeneity and treatment regimens and can be implemented from standard H&E slides. Experimental testing will involve retrospective discovery and external validation cohorts, with digital pathology pipelines to quantify sTIL% and spatial metrics, followed by multivariable modeling (including interaction terms) and assessment of discriminative performance (AUC, calibration).

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density (stromal TILs on H&E)
Primary endpoint	therapy_response

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

Prediction 3: In functional experiments, restoring or enhancing the anchor gene's signaling will improve immune-mediated tumor control in imaging-feature-positive models.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.
- Multi-modal validation pipeline: computational -> spatial -> functional, providing mechanistic depth beyond association.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions